

MD. SALMAN KHAN

Software Engineering Student

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Malek Ukil Hall, NSTU, Noakhali

OVERVIEW

Dynamic and motivated Software Engineering student with a strong foundation in computer science, software development, and problem-solving. Skilled in building scalable web applications and implementing efficient software solutions using modern technologies. Passionate about competitive programming, full-stack development, and system design, with the ability to quickly adapt to new technologies and collaborate effectively within development teams.

EDUCATION

Noakhali Science and Technology University	2019–2020
Bachelor of Science in Software Engineering (BSSE)	
Year-3 Term-2	CGPA: 3.61
Cumilla Victoria Govt. College	2021
Higher Secondary Certificate, Cumilla Board	GPA: 5.00
Debidwer Reaz Uddin Pilot Govt. High School	2019
Secondary School Certificate, Cumilla Board	GPA: 5.00

PROJECTS

YC-Interior	Ongoing
- Developed a modern full-stack interior design and architecture platform focused on premium UI/UX, cinematic animations, and responsive client-facing experiences. - Implemented dynamic landing pages, smooth scrolling interactions, animated statistics sections, staggered text animations, and centralized animation architecture using Angular services. - Designed scalable API-driven frontend/backend architecture with a strong emphasis on luxury aesthetics, motion design, and professional branding. - Technologies: Angular, Spring Boot, MySQL, REST API, Lenis. - GitHub: github.com/salman-nstu/YC-Interior	
MonProva	November 2025 – January 2026
- Developed a secure and scalable mental healthcare platform connecting patients, doctors, and administrators through a unified digital ecosystem. - Implemented clinically validated mental health self-assessments (PHQ-9, GAD-7, PSS-10), appointment booking, digital prescriptions, wellness activities, and real-time notifications. - Built role-based dashboards with JWT authentication, Firebase authentication, secure payment integration, and cloud-based document management. - Technologies: React 18, Node.js, Express.js, MongoDB Atlas, Firebase, JWT, Tailwind CSS, Cloudinary, SSLCommerz. - GitHub: github.com/salman-nstu/monprova	
SmartHeap - Thread-Safe Custom Memory Allocator	May 2025 – August 2025
- Developed a production-quality custom memory allocator in C implementing malloc(), free(), calloc(), and realloc() using native Windows APIs. - Implemented advanced memory management techniques including block splitting, coalescing, arena-based allocation, fragmentation analysis, corruption detection, and leak tracking.	

- Designed multiple allocation strategies (First Fit, Best Fit, Worst Fit) with thread-safe concurrent access using *CRITICAL_SECTION* synchronization.
- Technologies: C, Windows API, VirtualAlloc, Memory Management, Concurrency, Data Structures.
- GitHub: github.com/salman-nstu/SmartHeap

LectureLens - AI Powered Lecture Summarizer

January 2025 – March 2025

- Developed an AI-powered lecture summarization platform capable of transcribing and summarizing lectures from microphone input, uploaded media files, and YouTube videos.
- Integrated OpenAI Whisper for speech-to-text transcription and Facebook BART for intelligent abstractive summarization with export support for PDF, DOCX, and JSON formats.
- Implemented responsive frontend architecture with real-time feedback, dark/light themes, progress tracking, and REST API integration.
- Technologies: Python, Flask, OpenAI Whisper, Hugging Face Transformers, JavaScript, HTML, CSS, PyTorch.
- GitHub: github.com/salman-nstu/LectureLens

ProCoordinator

March 2024 – June 2024

- Developed a desktop-based Critical Path Method (CPM) Calculator to assist in project scheduling, dependency analysis, and workflow optimization.
- Implemented algorithms for calculating Earliest Start (ES), Earliest Finish (EF), Latest Start (LS), Latest Finish (LF), slack time, and identifying the critical path of complex projects.
- Designed an interactive Java GUI with color-coded visualization tables, automated result exporting, and task dependency management.
- Technologies: Java, Swing GUI, OOP, Algorithms, Data Structures.
- GitHub: github.com/salman-nstu/ProCoordinator

Password Manager with Multi-Layered Security

- Developed a secure password management application with encryption and hashing mechanisms.
- Implemented AES encryption and SHA-based hashing for secure credential storage.
- Technologies: Java, AES, SHA, MySQL.

IoT-Enabled Smart Farming System

- Designed an IoT-based smart farming solution for monitoring environmental conditions and crop management.
- Integrated sensor-based monitoring and automated notifications.
- Technologies: IoT, Embedded Systems, Sensors, Web Dashboard.

Java Banking System

- Developed a banking management application with authentication and account management features.
- Implemented login/signup system, transaction handling, and file-based credential verification.
- Technologies: Java, File Handling, OOP.

TECHNICAL SKILLS

Programming Languages	Java, C++, JavaScript, Python
Frontend	React, Angular, HTML, CSS
Backend	Spring Boot, Node.js
Databases	MySQL, MongoDB
Tools & Platforms	Git, GitHub, Firebase, Docker, Linux, VS Code
Core Areas	Competitive Programming, Full Stack Development, Problem Solving

COMPETITIVE PROGRAMMING & ONLINE JUDGES

Platform	Username / Profile	Achievement
Codeforces	CodeWarriorSK	Solved 200+ problems, Max Rating: 1100
LeetCode	warrior-salman	Solved 100+ problems
Beecrowd	Salman Khan	Solved 50+ problems

POSITIONS OF RESPONSIBILITY

- **Executive Member**, IT Club, NSTU 2024–Present
- **Programming Instructor**, Debidwer Publician Association 2025–Present
- **Program Secretary**, Youth Network Centre 2022–Present
- **Organizing Committee Member**, Mental Health Care 2020–Present

ACHIEVEMENTS & CERTIFICATIONS

SL	Name	Year	Description
1	2nd Runner-up, Programming Contest	2024	NSTU Intra University Programming Contest organized by CSTE Club.
2	2nd Place, SRS Poster Presentation	2024	Intra-University Software Project Showcasing organized by IT Club, NSTU.
3	6th Place, Programming Contest	2023	NSTU EEE DAY 2023 Inter University Programming Contest (Regional).

LANGUAGE PROFICIENCY

- English — Professional Working Proficiency
- Bengali — Native Speaker

REFERENCES

Md. Iftekhar Alam Efat

Assistant Professor

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